

ANGCHEN XIE

Robotics Institute, Carnegie Mellon University

xieangchen808@gmail.com | angchenxie.github.io | [Google Scholar](#) | [GitHub](#) | [X](#)

EDUCATION

Carnegie Mellon University

M.S. in Robotics (MSR)

Pittsburgh, PA

Aug. 2025 – May 2027 (expected)

- GPA: 4.17/4.00 | Advisors: [Guanya Shi](#), [Max Simchowitz](#)
- Core courses: 16-831 Introduction to Robot Learning; 16-745 Optimal Control and Reinforcement Learning; 16-720 Computer Vision

Shanghai Jiao Tong University

B.Eng. in Automation

Shanghai, China

Sept. 2021 – June 2025

- GPA: 4.01/4.30 | Rank: 1/87

RESEARCH INTERESTS

I work toward general-purpose robots that combine broad learned priors with grounded models of themselves and the physical world, enabling reliable and versatile behavior across tasks, embodiments, and environments. Three threads run through my work: adapting whole-body humanoid control to unseen dynamics from only minutes of target-domain data; controlling the rich, sustained contact that arises when a robot works on an object or a person; and building the infrastructure that lets policies and data move between robots. I am drawn to methods that have to survive contact with real hardware rather than only a simulator.

PUBLICATIONS

* Equal contribution.

- [1] **Angchen Xie***, Nikhil Sobanbabu*, Ishayu Shikhare, Alan Wang, Max Simchowitz, Guanya Shi
FADA: Few-Shot Domain Adaptation via Dynamics Alignment for Humanoid Control
Under review at CoRL [\[arXiv\]](#) [\[Website\]](#) [\[Thread\]](#)
- [2] Rishabh Madan, **Angchen Xie**, Samantha Saak, Andres Blanco, Dohyeok Lee, Sarah Grace Brown, Yunting Yan, Mark Zolotas, Jose Barreiros, Tapomayukh Bhattacharjee
TACTIC: Tactile and Vision Conditioned Contact-Centric Control for Whole-Arm Manipulation
Robotics: Science and Systems (RSS), 2026 [\[arXiv\]](#) [\[Website\]](#)
- [3] Pablo Ortega-Kral*, Eliot Xing*, Arthur Buckner, Vernon Luk, Junseo Kim, Owen Kwon, **Angchen Xie**, Nikhil Sobanbabu, Yifu Yuan, Megan Lee, Deepam Ameria, Bhaswanth Ayapilla, Jaycie Bussell, Guanya Shi, Jonathan Francis, Jean Oh
RIO: Flexible Real-Time Robot I/O for Cross-Embodiment Robot Learning
Robotics: Science and Systems (RSS), 2026 [\[arXiv\]](#) [\[Website\]](#) [\[Code\]](#)
- [4] Rishabh Madan, Jiawei Lin, Mahika Goel, **Angchen Xie**, Xiaoyu Liang, Marcus Lee, Justin Guo, Pranav N. Thakkar, Rohan Banerjee, Jose Barreiros, Kate Tsui, Tom Silver, Tapomayukh Bhattacharjee
PrioriTouch: Adapting to User Contact Preferences for Whole-Arm Physical Human-Robot Interaction
Conference on Robot Learning (CoRL), 2025 [\[arXiv\]](#) [\[Website\]](#)
- [5] **Angchen Xie**, Yeqiang Qian, Weihao Yan, Chunxiang Wang, Ming Yang
Non-repetitive: A Promising LiDAR Scanning Pattern
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024 [\[Paper\]](#) [\[Dataset\]](#)

RESEARCH EXPERIENCE

LeCAR Lab, Carnegie Mellon University

Research Intern

Pittsburgh, PA

Aug. 2025 – Present

- Advisors: Guanya Shi, Max Simchowitz
- Proposed a Planner-Inverse-Dynamics architecture for humanoid whole-body control that adapts to unseen terrain, payload, and actuator dynamics by finetuning only the inverse dynamics model on about two minutes of target-domain rollouts, without rewards or demonstrations. (*FADA*)
- Built and released open-source infrastructure for cross-embodiment robot learning, validated on vision-language-action deployment across three morphologies and four hardware platforms. (*RIO*)

RL² Lab, Shanghai Jiao Tong University

Research Intern

Shanghai, China

Jan. 2025 – June 2025

- Advisor: Yue Gao
- Applied model-based reinforcement learning to hexapod locomotion across varied terrain and deployed the resulting policies on hardware.
- Designed a fault-tolerant controller that adapts to the failure of one or more joints, including failure modes unseen during training.
- Introduced multi-sensor fusion to sharpen the robot's perception of terrain and of its own failure state, improving generalization on the real platform.

EmPRISE Lab, Cornell University

Research Intern

Ithaca, NY

July 2024 – Dec. 2024

- Advisor: Tapomayukh Bhattacharjee
- Developed a contact-centric predictive model for whole-arm manipulation that couples learned latent dynamics with analytical kinematics through contact Jacobians, and drove it with a contact-aware sampling-based MPC planner. (*TACTIC*)
- Contributed to a learning-to-rank framework that resolves conflicting user force preferences across simultaneous contacts in physical human-robot interaction. (*PrioriTouch*)

CyberC3 Intelligent Vehicle Lab, Shanghai Jiao Tong University

Research Intern

Shanghai, China

Nov. 2022 – June 2024

- Advisors: Yeqiang Qian, Ming Yang
- Built *Repetitive-or-not*, a CARLA dataset captured simultaneously by repetitive and non-repetitive LiDARs, isolating scanning pattern as the single variable.
- Showed that non-repetitive scanning is the more promising pattern on the strength of its object scanning, raising the maximum detection count on small traffic participants by 59.6%, and quantified the domain gap between the two point-cloud types. (*IROS 2024*)

SERVICE

Reviewer, Conference on Robot Learning (CoRL)

2026

Reviewer, Robotics: Science and Systems (RSS)

2026

Reviewer, IEEE Robotics and Automation Letters (RA-L)

2025, 2026

HONORS AND AWARDS

Outstanding Graduate Award, Shanghai Jiao Tong University

2025

China National Scholarship (top 0.2%)

2024

First Prize Academic Scholarship, Shanghai Jiao Tong University

2024

National First Prize, China Undergraduate Electronics Design Contest (top 1%)

2023

First Prize, China Undergraduate Mathematical Contest in Modeling (top 2%)

2022